

Formal Record of Dissertation Meeting

Student's Responsibility

It is the student's responsibility to take notes of the meeting and forward them to the supervisor in a timely manner.

Student:	Mykhailo Shpyl	Supervisor:	Dr. Vini Vijayan
-----------------	----------------	--------------------	------------------

Programme:	MSc Artificial Intelligence and Big Data Analytics
Date:	16 June 2026 (Meeting 3)

Key Points Discussed (must include timetable for research and training record):

Progress since Meeting 2. The matched AI-essay dataset is complete: 640 AI essays generated locally with Llama 3.1, matched to the human essays on topic and length, giving a 1,280-essay corpus. A first detector has been trained.

Detector audit. The first detector scored 100 percent, which was treated as a warning rather than a result. The human BAWE essays were found to carry structural export tags (headings, footnotes) that the AI essays do not, so a tag-only rule alone reaches 92.5 percent. After stripping the markup from both sides the detector still scores an F1 of 0.99, and a model using only function words reaches 99.5 percent, so the remaining signal is genuine writing style, not formatting or topic.

Explainability started. Showed the words that push a decision each way (human connectives versus the AI register) and the two clusters that essays form in style space. Explainability is one of the main research topics and is now beginning.

Why detection is easy at this stage. With only human writing and one AI model (Llama), the classes separate cleanly. As more generators (GPT, Claude, DeepSeek) are added the classes will overlap and need more care, and the dataset will need AI text that a human has rewritten or paraphrased, which would otherwise distort the results.

Scope of AI use. Many universities and journals now accept work modified with generative AI tools, so blanket detection is not the goal. The case of interest is AI-written text that a human has tweaked (the partial-AI category), which stays later-phase. The core scope remains two classes.

Language of the submission. The detector should account for the expected language variety of submissions (British English for ATU students) so that language locale is not mistaken for AI authorship; the AI generation should match that locale.

Compute. Confirmed that ATU HPC and cloud will not be available. The whole project runs on the laptop (RTX 4060, 8 GB), so every component must be designed to fit that budget.

Publication. AICS, the Irish Conference on Artificial Intelligence and Cognitive Science, is confirmed to be running, and the supervisor received an email about it. The submission window is near (historically a December conference with a deadline in September to October). GenAIDetect is a second target. The student is locating the call and the submission forms.

Materials and timetable. The drafted chapters and the presentation will be submitted now so progress is visible. Foundation and detection are done; next are the implementation write-up and explainability, then question generation and the rest of the pipeline, with writing throughout. About two and a half months remain to the end-of-August deadline.

Action items for Student:

Formal Record of Dissertation Meeting

- Submit the drafted chapters and the presentation as a separate file to the supervisor, and upload this record to OneDrive.
- Write the implementation chapters (dataset construction and generation, the detector, and the audit).
- Research the AICS application and proposal process (Student Track) and prepare a 6 to 12 page CEURART submission; keep GenAIDetect as a second target.
- Improve multi-generator detection and dataset consistency, and plan for paraphrased or partially human-edited AI text.
- Match the AI generation to British English so language locale is not learned instead of writing style.
- Continue the explainability work (the attribution methods) as the next main component.
- Clarify the application form mentioned previously and email the supervisor once the materials are uploaded.

Action items for Supervisor:

- Share the AICS 2026 call and proposal details received by email.
- Advise on the question-generation backend given the laptop-only constraint, since a full Llama 3 8B QLoRA may be too tight for 8 GB and a smaller open model may be needed.
- Review the submitted chapters and presentation before the next meeting if time allows.

Date of next meeting:	Tuesday 23 June 2026, 12:00 (online)
------------------------------	--------------------------------------

Issues requiring immediate attention:

AICS proposal window. The conference is confirmed and the deadline window is near, so the call must be found and a Student Track paper drafted soon.

Compute constraint. HPC and cloud are not available, so the project is laptop-only (8 GB). The question-generation backend (Llama 3 8B with QLoRA) may not fit, and a smaller open model may be needed. This touches the locked scope and needs supervisor sign-off before any change.

Dataset realism. Multi-generator data and paraphrased or partially human-edited AI text are the next dataset priorities, since the current single-generator setup makes detection easier than the real task.